

Assignment: Digital Forensics

1. Explain Forensics

Digital forensics is the process of identifying, collecting, preserving, examining, analyzing, and presenting digital evidence from electronic devices such as computers, mobile phones, servers, and networks. It helps investigators determine how a cyberattack happened, who was responsible, what data was affected, and how similar incidents can be prevented in the future.

Example:

Suppose a company's web server is hacked and customer data is stolen. A digital forensic investigator examines the server logs, network traffic, and system files to determine how the attacker gained access, what actions they performed, and which files were accessed or stolen.

2. List Evidence

Digital evidence is any information stored or transmitted in digital form that can be used during an investigation.

Some common types of digital evidence include:

- **System Logs** – Records of user logins, software activity, and system events.
 - *Example:* Windows Event Logs showing repeated failed login attempts.
- **Hard Drive Images** – An exact copy of a computer's storage device.
 - *Example:* A forensic image of a suspect's laptop used to recover deleted files.
- **Memory (RAM) Dumps** – Data stored in memory while a system is running.
 - *Example:* Recovering malware that only exists in memory.
- **Network Traffic (Packet Captures)** – Data exchanged between devices over a network.
 - *Example:* Capturing suspicious outbound traffic that indicates data exfiltration.
- **Emails and Attachments**
 - *Example:* A phishing email containing a malicious attachment used to infect a system.
- **Browser History**
 - *Example:* Finding that a user visited a malicious website before malware was installed.
- **USB Device History**
 - *Example:* Evidence showing that confidential files were copied to a USB drive.
- **Deleted Files and File Metadata**
 - *Example:* Recovering deleted documents and identifying when they were created or modified.

3. Describe Steps

A digital forensics investigation usually follows these steps:

Step 1: Identification

Identify the devices, accounts, and data sources that may contain evidence.

Example: Investigators identify the compromised web server and the employee's workstation for examination.

Step 2: Preservation

Protect the evidence from being altered by creating forensic copies and maintaining a chain of custody.

Example: A forensic image of the server's hard drive is created before any analysis begins.

Step 3: Collection

Collect digital evidence using forensic tools while ensuring data integrity.

Example: Collecting server logs, memory dumps, email records, and firewall logs.

Step 4: Examination

Extract and organize relevant information from the collected evidence.

Example: Recovering deleted files and reviewing browser history for suspicious activity.

Step 5: Analysis

Analyze the evidence to determine the attack method, timeline, and impact.

Example: The investigator discovers that the attacker exploited a web application vulnerability, installed malware, and stole customer records.

Step 6: Reporting

Prepare a detailed report documenting the investigation, findings, evidence collected, and recommendations.

Example: The report explains how the attack occurred, identifies the compromised accounts, estimates the data affected, and recommends implementing multi-factor authentication and regular security updates.